STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 17

PROBABILITY DISTRIBUTIONS

UNIFORM DISTRIBUTION

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UNIFORM DISRIBUTION

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THE UNIFORM DISTRIBUTION

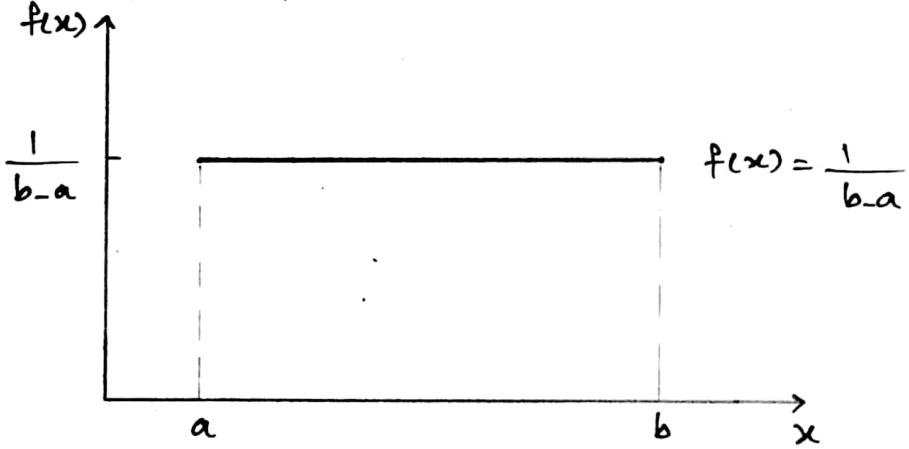
INTRODUCTION The term uniform distribution was first introduced by J. V. Uspensky in Introduction to Mathematical Probability. A uniform random variable has the property that it gives equal chances to all the members of the population. In practice, the distribution is generally used when every point in an interval is equally likely to occur, or at least insufficient knowledge exists to propose another model.

DEFINITION A random variable 'x' is said to be uniformly distributed over the interval (a, b) if its probability density function is given by

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise.} \end{cases}$$

ie "A stochastic variable is said to have uniform distribution of probability if probabilities attached to two equal intervals are equal."

Graph of p.d.f of uniform distribution is given below

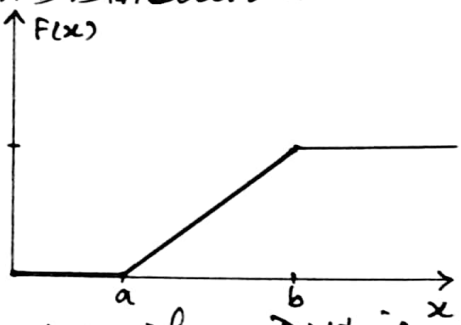


uniform distribution is also called rectangular distribution. because its total probability is confined to a rectangular region with base $(b-a)$ and height $\frac{1}{b-a}$.

PROPERTIES OF UNIFORM DISTRIBUTION

1. The shape of uniform distribution is rectangular.
2. Uniform Distribution has two parameters 'a' and 'b'.
3. Mean of uniform distribution $= \frac{a+b}{2}$
4. Variance of uniform distribution $= \frac{(a-b)^2}{12}$
5. Distribution Function of uniform Distribution is given by

$$F(x) = \begin{cases} 0 & , \ x < a \\ \frac{x-a}{b-a} & , \ a \le x \le b \\ 1 & , \ x > b \end{cases}$$



6. Moment Generating Function of uniform Dist is given by

$$M_x(t) = \frac{e^{bt} - e^{at}}{(b-a).t}$$

7. Total Area under the uniform Random variable is unity. ie $\int_a^b f(x)\,dx = 1$

8. The rth raw Moment of uniform Random variable is

$$E(X^r) = \frac{1}{(b-a)(r+1)} \left[b^{r+1} - a^{r+1} \right]$$

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EXAMPLES OF UNIFORM DISTRIBUTION

EXAMPLE-① If x is uniformly distributed over $(0,10)$, calculate the probability that

(a) $P(x<3)$

(b) $P(x>6)$

(c) $P(3<x<8)$

SOLUTION For the interval $(0,10)$, $f(x)=\frac{1}{10-0}=\frac{1}{10}$

Now

(a) $P(x<3)=\int_0^3 \frac{1}{10}dx=\frac{1}{10}|x|_0^3=\frac{3}{10}$

(b) $P(x>6)=\int_6^{10} \frac{1}{10}dx=\frac{1}{10}|x|_6^{10}=\frac{4}{10}$

(c) $P(3<x<8)=\int_3^8 \frac{1}{10}dx=\frac{1}{10}|x|_3^8=\frac{1}{10}\times 5=\frac{1}{2}$

EXAMPLE② Suppose that a large conference room at a certain company can be reserved for no more than 4 hours. Both short and long conferences occur quite often. In fact, it can be assumed that the length x of conference has a uniform distribution on the interval $[0,4]$,

(a) what is the Probability density function.

(b) what is the Probability that any given conference lasts at least 3 hours.

SOLUTION (a) On this situation appropriate p.d.f is given by

$$f(x)=\begin{cases}\frac{1}{4}, & 0\le x\le 4\\ 0, & \text{otherwise.}\end{cases}$$

(b) $P(x\geq 3)=\int_3^4 \frac{1}{4}dx=\frac{1}{4}|x|_3^4=\frac{1}{4}$

EXAMPLE-③ Buses arrive at a specified stop at 15-minute intervals starting at 7 A.M. That is, they arrive at 7, 7:15, 7:30, 7:45 and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7 and 7:30, find the probability that he waits.

(a) less than 5 minutes for a Bus.

(b) more than 10 minutes for a Bus.

SOLUTION Let 'x' denote the number of minutes past 7 that the passenger arrives at the stop. Since 'x' is uniformly distributed over the interval (0,30), so

$$f(x) = \begin{cases} \frac{1}{30}, & 0 \le x \le 30 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Passengers will have to wait less than 5 minutes if he arrives b/w 7:10 and 7:15 or between 7:25 and 7:30.

Hence $P(10 < X < 15) + P(25 < X < 30)$

$$= \int_{10}^{15} \frac{1}{30} dx + \int_{25}^{30} \frac{1}{30} dx$$

$$= \frac{1}{30}|x|_{10}^{15} + \frac{1}{30}|x|_{25}^{30} = \frac{1}{30}(5) + \frac{1}{30}(5) = \frac{1}{3}$$

(b) Passenger will have to wait more than 10 minutes if he arrives between 7 and 7:05 or between 7:15 and 7:20. So

$P(0 < X < 5) + P(15 < X < 20)$

$$= \int_0^5 \frac{1}{30} dx + \int_{15}^{20} \frac{1}{30} dx = \frac{1}{30}(5) + \frac{1}{30}(5) = \frac{1}{3}$$

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EXAMPLE-④ A farmer living in Siaalkot has an irrigation system to provide water for crops, primarily corn, on a large farm. Although he has thought about buying a backup pump, he has not done so. If the pump fails, delivery time for a new pump to arrive is uniformly distributed over the interval from 1 to 4 days. The pump fails. It is critical time in the growing season in that the yield will be greatly reduced if the crop is not watered with in the next 3 days. Assuming that the pump is ordered immediately and that installation time is negligible, what is the probability that farmer will suffer major yield loss?

SOLUTION.

Let T be the time until the pump is delivered. T is uniformly distributed over the interval (1,4).

$$\text{So } f(t) = \begin{cases} \frac{1}{3}, & 1 \le t \le 4 \\ 0, & \text{otherwise.} \end{cases}$$

We have to find $P(T>3) = ?$

$$\text{So } P(T>3) = \int_3^4 f(t)\,dt$$

$$= \int_3^4 \frac{1}{3}\,dt$$

$$= \frac{1}{3}|t|_3^4 = \frac{1}{3}(4-3) = \frac{1}{3}$$

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EXAMPLE-⑤ Let the continuous Random Variable 'x' denote the current measured in a thin copper wire in milliamperes. Assume the range of 'x' is [0, 20] and assume that the probability density function of x is

$$f(x) = \begin{cases} 0.05, & 0 \le x \le 20 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the Probability that a measurement of current is between 5 and 10 milliamperes.

(b) Also find E(x), V(x) and F(x).

SOLUTION Here $f(x) = \begin{cases} 0.05, & 0 \le x \le 20 \\ 0, & \text{otherwise.} \end{cases}$

(a) $P(5 < x < 10) = \int_5^{10} 0.05\,dx$

$$= (0.05)\,|x|_5^{10} = (0.05)(5) = 0.25$$

(b) $E(x) = \frac{a+b}{2} = \frac{0+20}{2} = 10\ mA.$

$$V(x) = \frac{(a-b)^2}{12} = \frac{(0-20)^2}{12} = \frac{400}{12} = 33.33\ mA^2$$

$$F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \le x \le b \\ 1, & x > b \end{cases}$$

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{20}, & 0 \le x \le 20 \\ 1, & x > 20 \end{cases}$$

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PROOFS OF IMPORTANT PROPERTIES

PROPERTY-① Total Area under the uniform RV is 1.

SOLUTION we have to show that for

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise} \end{cases}$$

$$\int_a^b f(x)\,dx = 1$$

$$LHS = \int_a^b f(x)\,dx$$

$$= \int_a^b \frac{1}{b-a}\,dx = \frac{1}{(b-a)} \left| x \right|_a^b = \frac{1}{(b-a)}(b-a) = 1 = RHS.$$

PROPERTY-② Mean of uniform Distribution is $\frac{a+b}{2}$

ie $E(x) = \frac{a+b}{2}$

SOLUTION $E(x) = \int_a^b x \cdot f(x)\,dx$

$$= \int_a^b x \cdot \frac{1}{b-a}\,dx$$

$$= \frac{1}{(b-a)} \cdot \left| \frac{x^2}{2} \right|_a^b$$

$$= \frac{1}{2(b-a)} \cdot (b^2 - a^2)$$

$$= \frac{1}{2(b-a)} \cdot (b-a)(b+a)$$

$$E(x) = \frac{a+b}{2}$$

PROPERTY-③ Variance of a uniform RV is $\frac{(a-b)^2}{12}$ or $\frac{(b-a)^2}{12}$

SOLUTION $V(X) = E(X^2) - (E(X))^2 \longrightarrow ①$

Now $E(X^2) = \int_a^b x^2 f(x)\,dx$

$$= \int_a^b x^2 \cdot \frac{1}{b-a}\,dx$$

$$= \frac{1}{(b-a)} \cdot \left| \frac{x^3}{3} \right|_a^b$$

$$= \frac{1}{3(b-a)} \cdot (b^3 - a^3)$$

$$= \frac{1}{3(b-a)} \cdot (b-a)(b^2 + a^2 + ab)$$

$$E(X^2) = \frac{b^2 + a^2 + ab}{3}, \quad \text{Also} \quad E(X) = \frac{a+b}{2}$$

$$\Rightarrow V(X) = \frac{a^2 + b^2 + ab}{3} - \left(\frac{a+b}{2}\right)^2$$

$$// = \frac{a^2 + b^2 + ab}{3} - \frac{a^2 + b^2 + 2ab}{4}$$

$$// = \frac{4a^2 + 4b^2 + 4ab - 3a^2 - 3b^2 - 6ab}{12}$$

$$// = \frac{a^2 + b^2 - 2ab}{12}$$

$$V(X) = \frac{(a-b)^2}{12}$$

$$V(X) = \frac{(b-a)^2}{12}$$

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PROPERTY-④ CDF of uniform RV is given by

$$F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \le x \le b \\ 1, & x > b \end{cases}$$

SOLUTION

$$\text{Now } f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise.} \end{cases}$$

Now

$$F(x) = \int_{-\infty}^{x} f(x)\,dx$$

$$= \int_{a}^{x} \frac{1}{b-a}\,dx$$

$$= \frac{1}{(b-a)} \cdot |x|_a^x$$

$$F(x) = \frac{x-a}{b-a}$$

$$\text{So } F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \le x \le b \\ 1, & x > b \end{cases}$$

PROPERTY-⑤ MGF of uniform RV is $\frac{e^{bt} - e^{at}}{t(b-a)}$.

SOLUTION $M_x(t) = E(e^{tx})$

$$= \int_a^b e^{tx} \cdot f(x)\,dx = \int_a^b e^{tx} \cdot \frac{1}{(b-a)}\,dx$$

$$= \frac{1}{(b-a)} \cdot \left| \frac{e^{tx}}{t} \right|_a^b = \frac{1}{t(b-a)} \cdot (e^{bt} - e^{at})$$

$$= \frac{e^{bt} - e^{at}}{(b-a).t.}$$

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